

Performance characterization of PET detectors and systems

Oscilloscope baseline, Pico-TDC performance, and waveform-ML correction

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Why timing matters in TOF-PET

TOF-PET imaging

A positron annihilation produces two photons emitted in opposite directions. Two detectors measure the corresponding signals and estimate the **difference in photon arrival time (TOF)**.

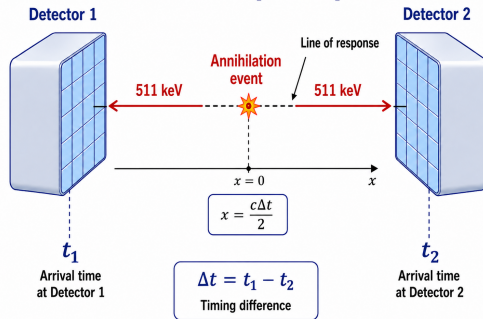
Why better timing helps

The arrival-time difference constrains the annihilation position along the line connecting the two detectors:

$$\Delta x = \frac{c \Delta t}{2}.$$

A more precise estimate of Δt means a more precise localization of the event.

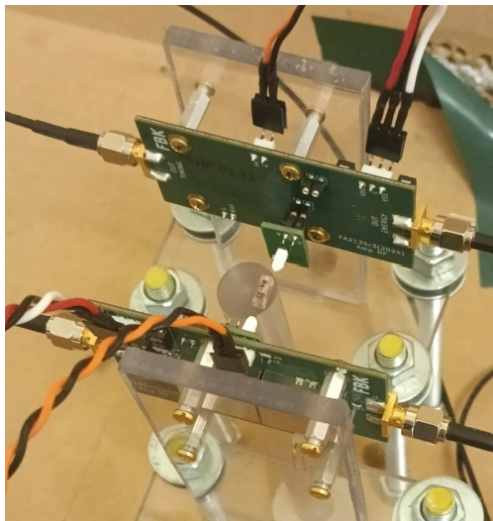
TOF-PET principle



Project question

What is the best timing performance achievable, and how can we approach it with a practical and scalable method?

Experimental setup



Detector system

- Two opposing LYSO–SiPM detector modules.
- LYSO crystal: $2 \times 2 \times 3 \text{ mm}^3$.
- BC SiPM: $4 \times 4 \text{ mm}^2$.
- FBK-HF front-end / amplifier.
- ^{22}Na positron-emitting source centered between the two detectors.

Analog outputs

For both detectors:

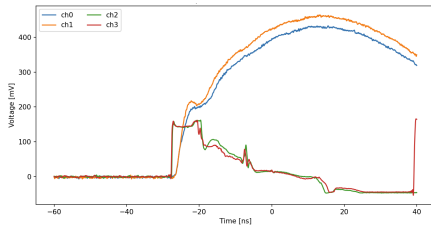
- **energy** channel: first stage of amplification.
- **timing** channel: second stage of amplification, designed for timing.

Two ways to acquire the detector signal

Oscilloscope: Full waveform

The voltage signal is sampled as a function of time.

- complete signal shape available offline;
- flexible timing analysis;



Pico-TDC: Threshold-based

Records the times at which the signal crosses a fixed threshold.

- leading and trailing crossing times;
- Time-over-Threshold (ToT);

```
//*****  
// File Format Version 3.3  
// Janus_5203 Release 3.0.0  
// Acquisition Mode: Trigger Matching  
// Measurement Mode: Lead Trail  
// ToA_LSB = 3.125 ps; ToT_LSB = 3.125 ps; Tstamp_LSB = 12.8 ns  
// Run50 start time: Wed Jul 1 21:06:25 2026 UTC  
//*****  
Tstamp_LSB   TrgID   Brd   Ch   E   ToA_LSB   ToT_LSB  
3315422      0      00   01   L   282084    -  
              00   01   T   306087    24003  
              00   03   L   65599     -  
              00   03   T   66039     440  
              00   03   L   113212    537  
              00   03   T   113775    48176  
              00   03   L   304133    644
```

Same detector signals, two different levels of information: full waveforms versus a small set of timestamps.

Why use the Pico-TDC?



Device

CAEN DT5203: 64-channel picoTDC unit for high-resolution ToA/ToT measurements (3.125 ps LSB).

Practical motivation

- **Lower cost** than high-bandwidth waveform digitization.
- **More channels** available in a compact system.
- Much **smaller data volume** per event.

Main question

Can this readout preserve the timing performance of full waveform acquisition?

Experimental data

Oscilloscope

- **Bias voltage:** 45–49 V, 1 V steps
- **Acquisition:** 100 ns waveforms at 80 GS/s

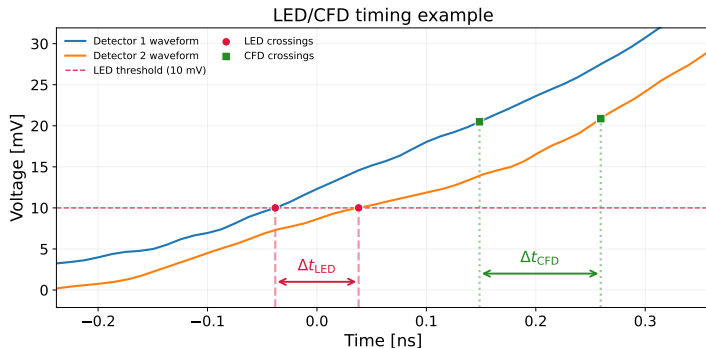
Bias [V]	Raw events	Photopeak events
45	20 135	6 763
46	21 510	8 037
47	20 073	8 482
48	20 264	7 049
49	25 054	7 078

Pico-TDC

- **Bias voltage:** 42–49 V, 1 V steps
- **Timing threshold:** 20–70 mV, 10mV steps
- **Acquisition:** trigger-matching mode

Bias [V]	T_{th} [mV]	Raw events	Photopeak events
42	40	79 127	11 514
43	40	84 586	11 470
44	40	90 229	11 554
45	40	96 008	11 552
45	40	96 001	11 658
46	20	49 348	5 793
46	30	49 688	5 909
46	40	100 916	11 847
46	50	50 245	5 931
46	60	50 271	5 800
46	70	50 410	5 981
47	40	104 916	11 881
48	40	107 613	12 182
49	40	111 406	12 434

Standard methods for oscilloscope waveforms



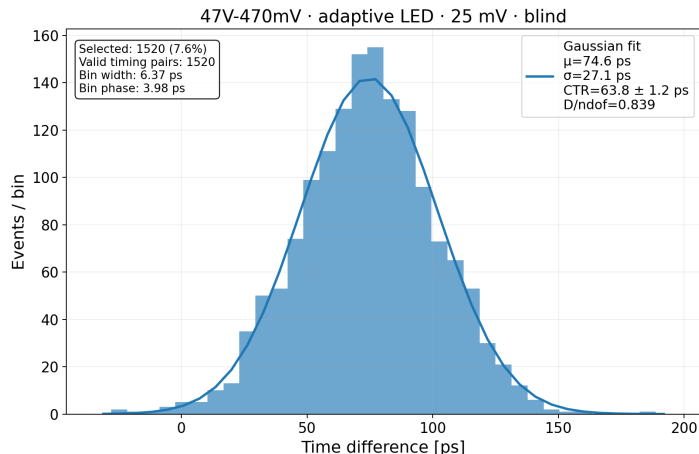
Leading-edge discriminator (LED)

$\Delta t_{LED} = t_{LED,1} - t_{LED,2}$ Fixed voltage crossing: simple and robust, but sensitive to amplitude-dependent time walk.

Constant-fraction discriminator (CFD)

$\Delta t_{CFD} = t_{CFD,1} - t_{CFD,2}$ Fractional crossing relative to pulse amplitude: reduces time walk, but still compresses the waveform to one time marker.

CTR extraction



Timing difference

For each selected coincidence event:

$$\Delta t = t_1 - t_2.$$

The selected events form the coincidence time distribution.

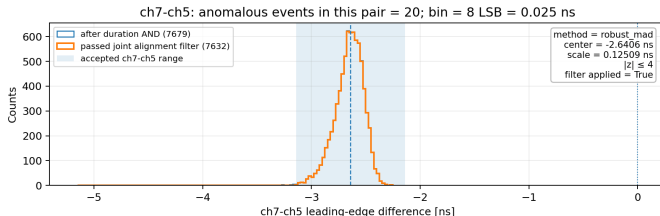
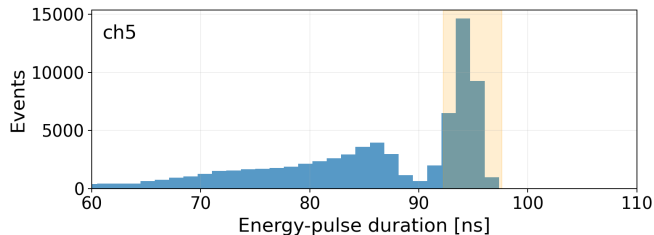
Coincidence Time Resolution (CTR)

$$\text{CTR} = \text{FWHM} = 2\sqrt{2 \ln 2} \sigma \simeq 2.35 \sigma.$$

Lower CTR means better TOF-PET image quality.

The same Gaussian-fit CTR definition is used for every timing method.

Pico-TDC analysis: from hits to coincidences



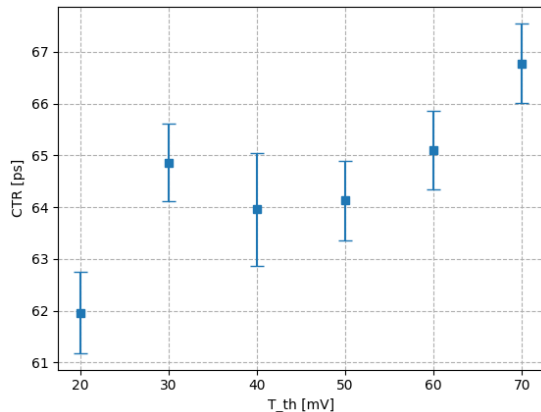
Pipeline

- 1 Raw timestamp lists
- 2 Leading/trailing pairs
- 3 ToT-based photopeak selection
- 4 Timing-hit matching
- 5 Mismatch rejection
- 6 Gaussian CTR fit

Personal contribution

Causal signal matching and filtering strategy to reject mismatched or noisy events while keeping lower timing thresholds usable.

Pico-TDC threshold scan



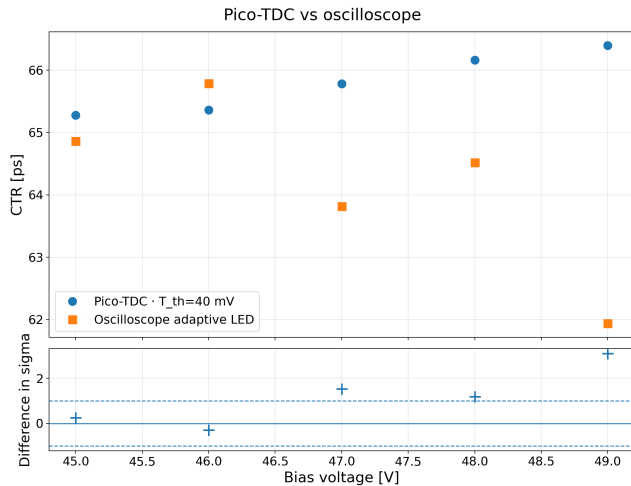
Threshold trade-off

- **Low threshold:** earlier pick-off, but more noise/spurious hits.
- **High threshold:** cleaner crossing, but more time walk and possible efficiency loss.

Analysis result

The threshold can affect significantly the performance and should be optimized.

Pico-TDC versus oscilloscope reference



Comparison

Error bars show the bootstrap uncertainty of the final fit only. They therefore underestimate the total analysis uncertainty.

Result

Using the timing channels, Pico-TDC can produce **CTR compatible with the oscilloscope reference**.

Takeaway

Pico-TDC is feasible for practical timing readout; the oscilloscope remains the baseline for studying the information limit.

Second question: can we do better?

Idea

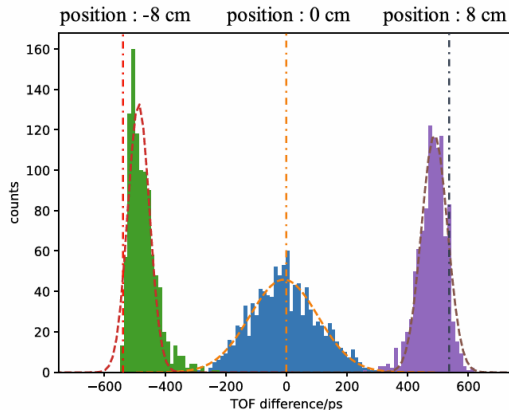
The signal shape contains information about how the event was measured.

Main question

Can a model learn a waveform-dependent timing correction, rather than a black-box TOF prediction?

Risk with Machine Learning

A very flexible model can make the time distribution narrower without necessarily learning the physical timing correction we want.



Source: X. Feng, A. Muhashi, Y. Onishi, R. Ota, and H. Liu, "Transformer-CNN hybrid network for improving PET time of flight prediction," *Physics in Medicine & Biology*, 2024.

What should the model learn?

Start from standard leading-edge timing

For each coincidence event, the measured time difference is: $\Delta t_{\text{LED}} = t_{\text{LED},1} - t_{\text{LED},2}$.

My modeling proposal

$$\Delta t_{\text{LED}} = \underbrace{\text{TOF}}_{\text{true time of flight}} + \underbrace{C_{1,2}}_{\text{calibration bias}} + \underbrace{\delta(s_1, s_2)}_{\text{waveform-dependent error}} + \underbrace{\epsilon}_{\text{random noise}}$$

ML target

$$\delta(s_1, s_2)$$

Where (s_1, s_2) is a signal pair.

Derived properties

- The physical TOF remains outside the ML model.
- Fixed channel offsets are handled by calibration.
- ML is used to correct waveform-dependent timing effects.

Physically constrained correction

Proposed model architecture: same model for all signals

$$y_{\theta}(s_1, s_2) = g_{\theta}(s_1) - g_{\theta}(s_2)$$

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \left(y_{\theta}(s_{1,i}, s_{2,i}) - (\Delta t_{\text{LED}} - \hat{C}_{1,2} - \text{TOF}) \right)^2$$

Corrected timing estimate

$$\hat{\Delta t} = \Delta t_{\text{LED}} - \hat{C}_{1,2} - y_{\theta}(s_1, s_2)$$

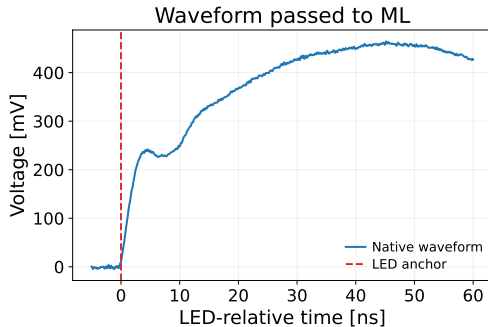
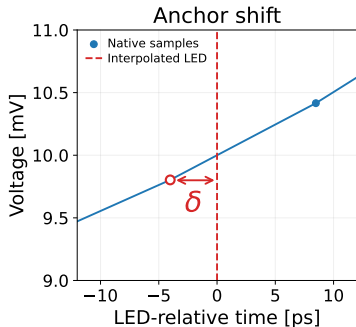
The final output is still a corrected arrival-time difference, so it can be evaluated with the same CTR metric.

Why this is safer than generic pair-ML

- Same pulse scorer for every detector.
- Learns a timing-error correction, not an absolute position label.
- No need for multiple source positions: a single central source is sufficient.

Constraining the model gives a translation-invariant, antisymmetric, and more general timing correction.

Windowing and target correction



Windowing

LED time is interpolated, but waveform samples remain on the native grid:

$$\delta_j = t_{\text{LED},j} - t_{a,j}.$$

Target

Correct for the discrete anchor shift:

$$y_{\text{target}} = \Delta t_{\text{LED}} - \Delta \delta - \text{TOF}$$

Analysis workflow: from detector data to result

One reproducible chain

Raw data → event selection → LED/CFD timing → ML correction → blind CTR → interpretation

1. Clean events

- Select photopeak events.
- Reject mismatched timing hits.
- Freeze standard LED/CFD references.

2. Compare readouts

- Full oscilloscope waveform.
- Pico-TDC timestamp readout.
- Reduced fixed-threshold representation.

3. Test fairly

- Tune on development data.
- Retrain selected configuration.
- Report one independent blind CTR.

Repository: `tof-pet-timing-uchicago-ONAOSI-prj3`

Models as controlled information probes

Machine learning models

Model	Description	Scientific question
Linear SVR	Weighted sum of waveform samples	Is the correction mostly linear?
CNN	Learns local waveform patterns	Does nonlinear waveform structure contain additional timing information?
Fixed-threshold SVR	Uses only a few crossing times	How much timing information survives a TDC-like readout?

Performance

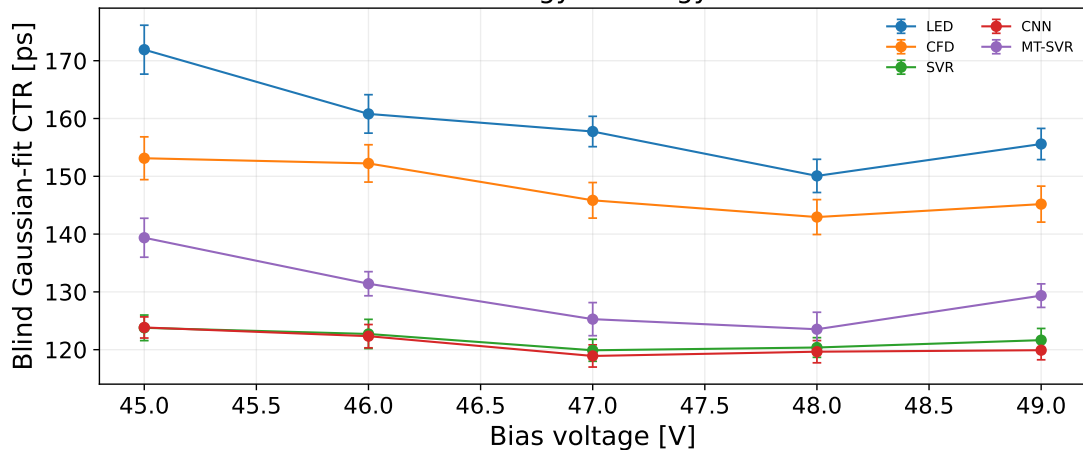
A model is useful if it reduces CTR on the independent blind set.

Interpretation

If different models learn similar event-by-event corrections, performance is likely limited by the information in the signal, rather than by model complexity.

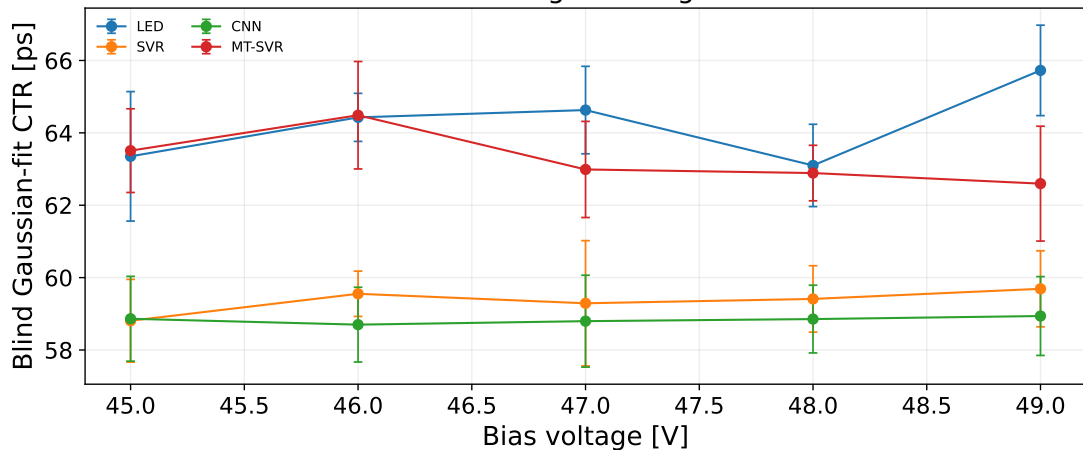
Waveform-ML results: energy-channel timing

energy → energy

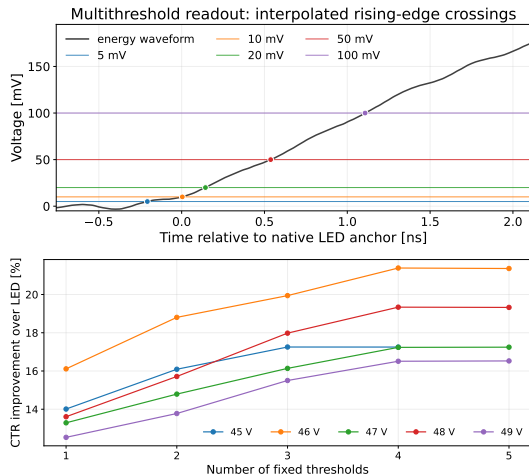


Waveform-ML results: timing-channel timing

timing → timing



Fixed-threshold timing as reduced readout



Reduced feature vector

$$\tau_k = t_1(V_k) - t_2(V_k), \quad x = [\tau_{k_1}, \dots, \tau_{k_m}].$$

Only m signal crossing times are passed to SVR.

Result

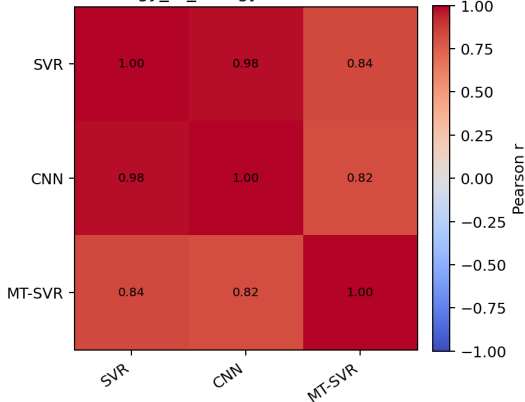
On energy channels, the first few thresholds recover much of the correction and can outperform standard single-time estimators.

Connection to Pico-TDC

This tests how much information can survive a TDC-like representation, without considering the ToT information.

XAI: do different models find the same correction?

46V-440mV · energy_to_energy · blind corrections · n=1607



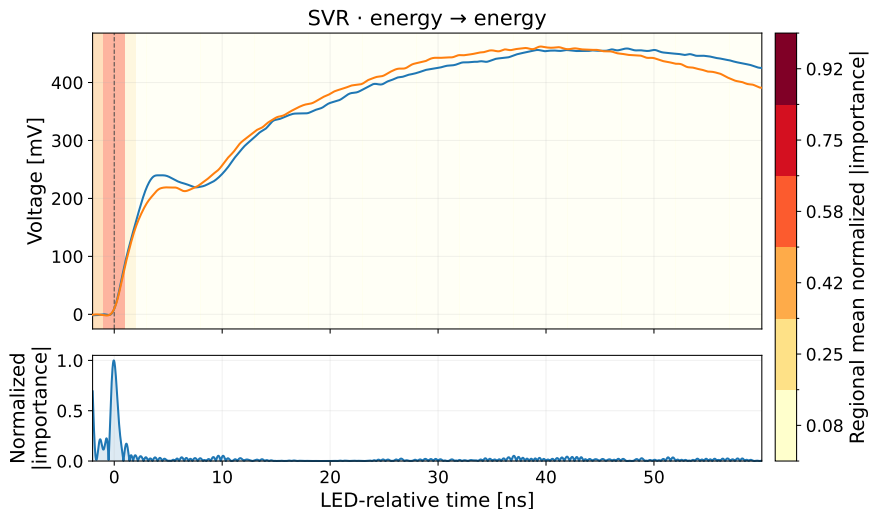
Result

SVR, CNN, and fixed-threshold SVR produce **strongly correlated event-by-event corrections**.

Interpretation

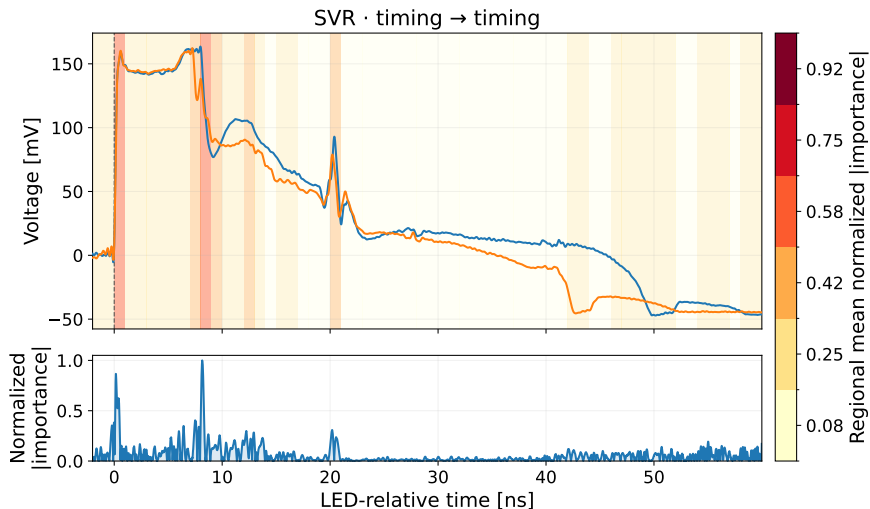
Similar corrections across different models suggest that signal information, not model complexity, limits performance.

XAI: energy-channel information



Most of the useful timing information is concentrated close to the signal arrival

XAI: timing-channel information



Useful regions remain more than 20 ns after signal arrival.

Conclusions

- Using the dedicated timing channels, Pico-TDC measurements achieve CTR compatible with the oscilloscope reference.
- ML corrects the waveform-dependent timing error, with the largest improvement observed for the energy channels.
- The shared antisymmetric model gives a translation-invariant correction, while a few threshold-crossing times retain much of the useful information.
- Timing-relevant information can persist after signal arrival, potentially suggesting alternative timing strategies and electronics designs.

Machine learning can improve timing and reveal where useful information is encoded in the detector signal.

Thank you for your attention.

References

- E. Berg and S. R. Cherry, "Using convolutional neural networks to estimate time-of-flight from PET detector waveforms," *Physics in Medicine & Biology*, 2018.
- X. Feng, A. Muhashi, Y. Onishi, R. Ota, and H. Liu, "Transformer-CNN hybrid network for improving PET time of flight prediction," *Physics in Medicine & Biology*, 2024.
- S. Naunheim et al., "Explicit machine-learning corrections and spatial generalization for PET detector timing," *Frontiers in Physics*, 2025.
- L. Y. Chang et al., "A Time-Walk Correction Method for PET Detectors Based on Leading Edge Discriminators," *IEEE Transactions on Radiation and Plasma Medical Sciences*, 2017.

Repository

tof-pet-timing-uchicago-ONAOSI-prj3